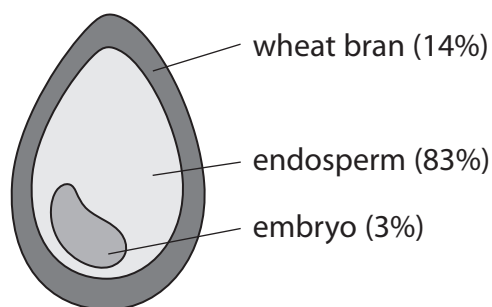


- 4 Plant-based products can replace plastics produced from oil.

Wheat bran comes from the outer layers of grains of wheat.

- (a) The diagram shows the structure and composition of a grain of wheat.



Wheat bran contains 43% fibre.

A grain of wheat has a mass of 48 mg.

Calculate the mass of fibre in the wheat bran of this grain of wheat.

Give your answer to **two** significant figures.

(3)

Answer mg

- (b) Disposable plates and cutlery can be made from either plant-based products or from oil-based plastics.

The photograph shows cutlery and a plate made from wheat bran.



(Source: © Konektus Photo/Shutterstock)

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Explain the advantages of using cutlery and plates made from wheat bran instead of oil-based plastics.

(3)

- (c) The endosperm of a grain of wheat develops from the endosperm nucleus, formed during fertilisation.

Explain the role of the pollen tube and nuclei in the formation of the endosperm nucleus.

(3)

(Total for Question 4 = 9 marks)

